

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

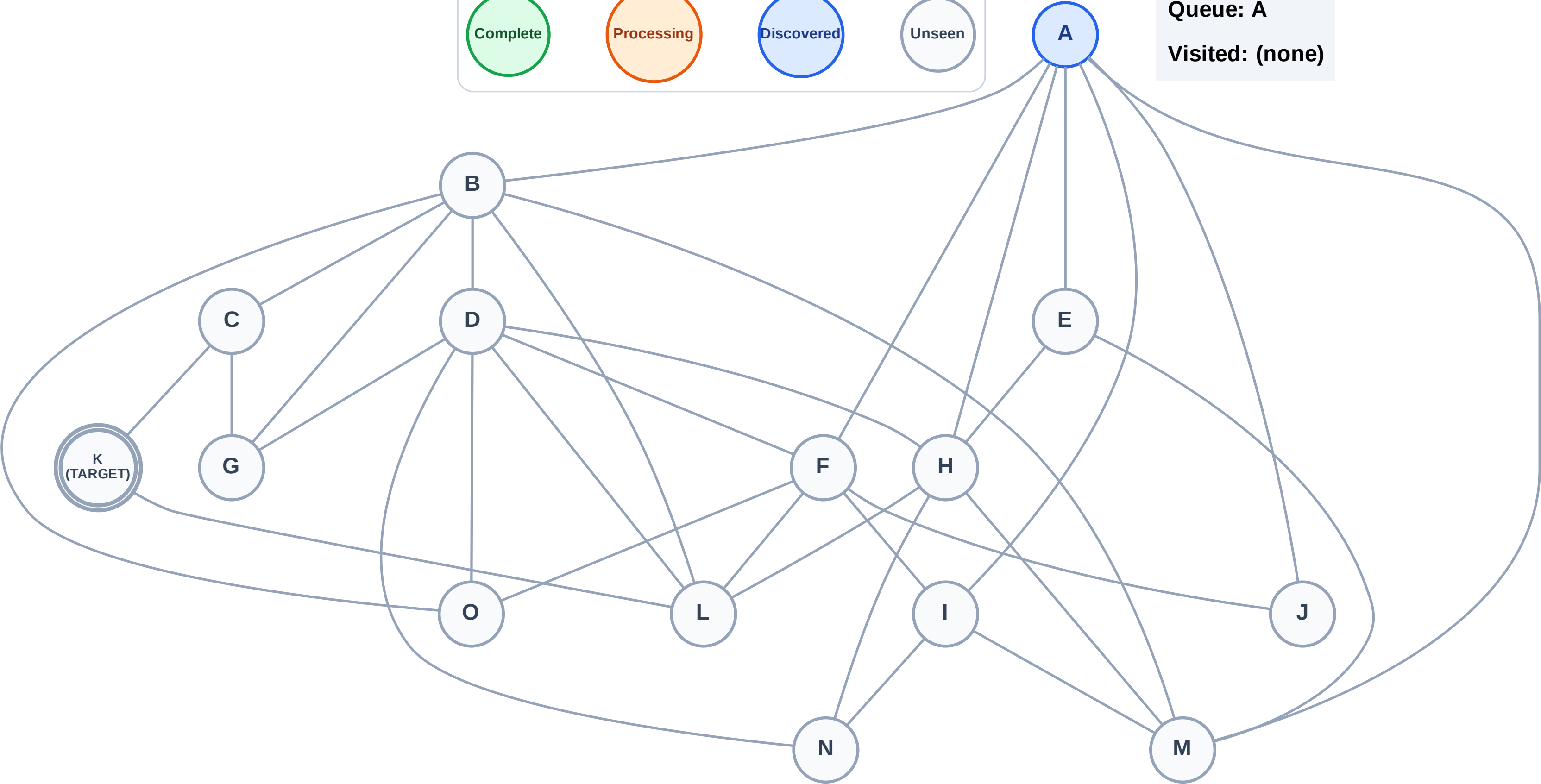
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

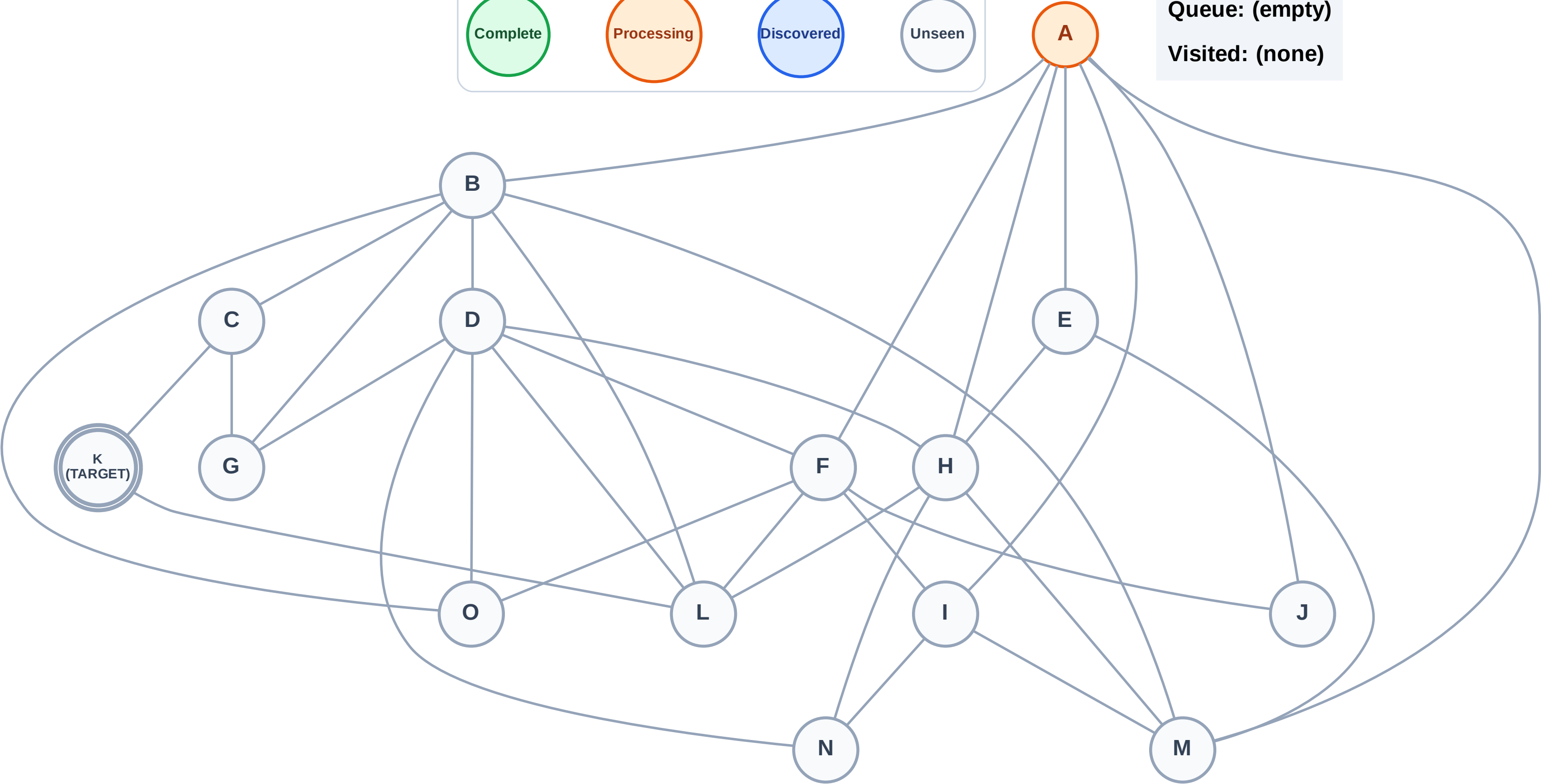
Complete

Processing

Discovered

Unseen

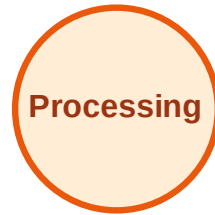
Queue: (empty)
Visited: (none)



BFS Traversal

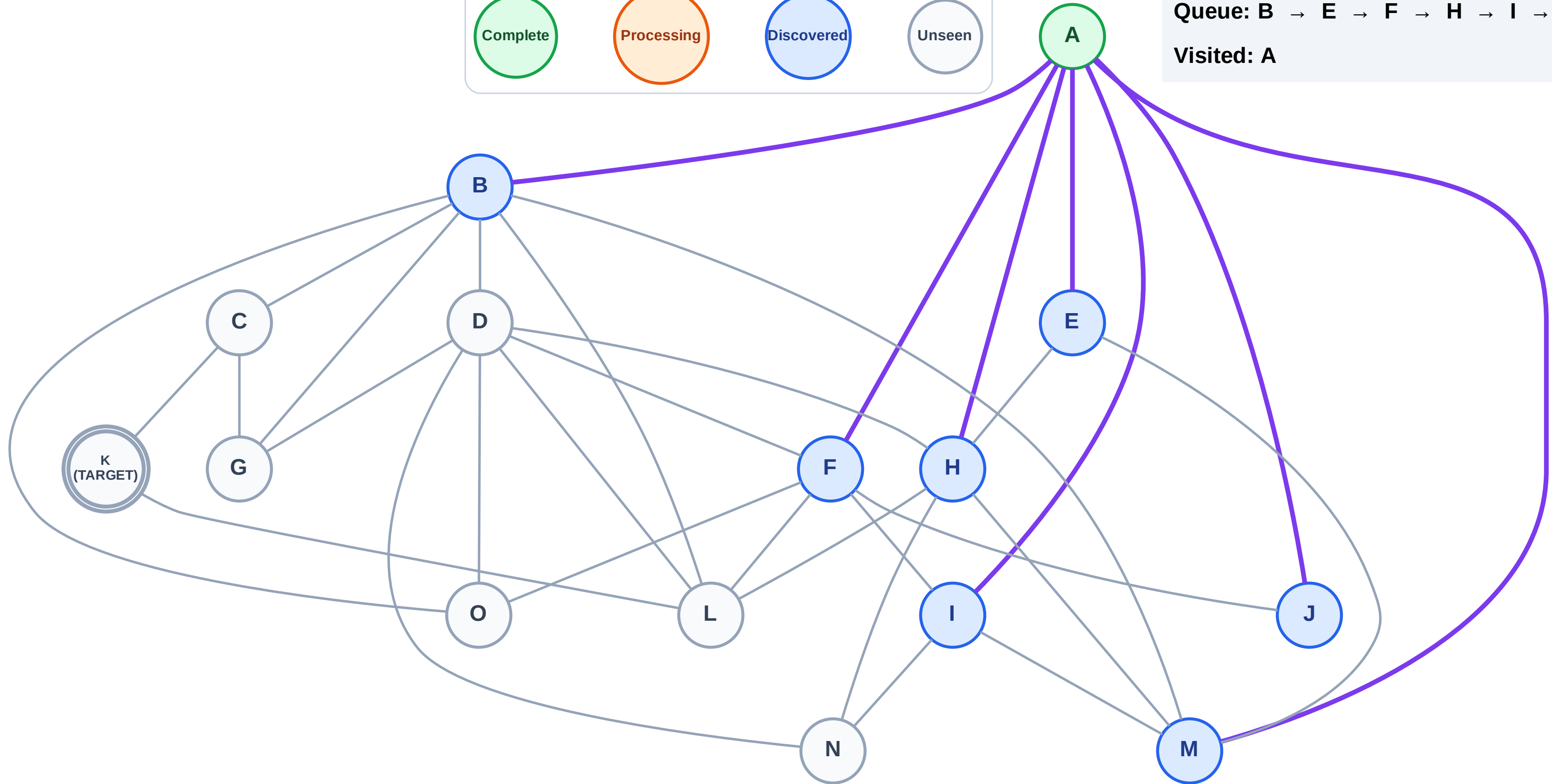
Step 3: Discover B, E, F, H, I, J, M from A

Legend



Queue: B → E → F → H → I → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

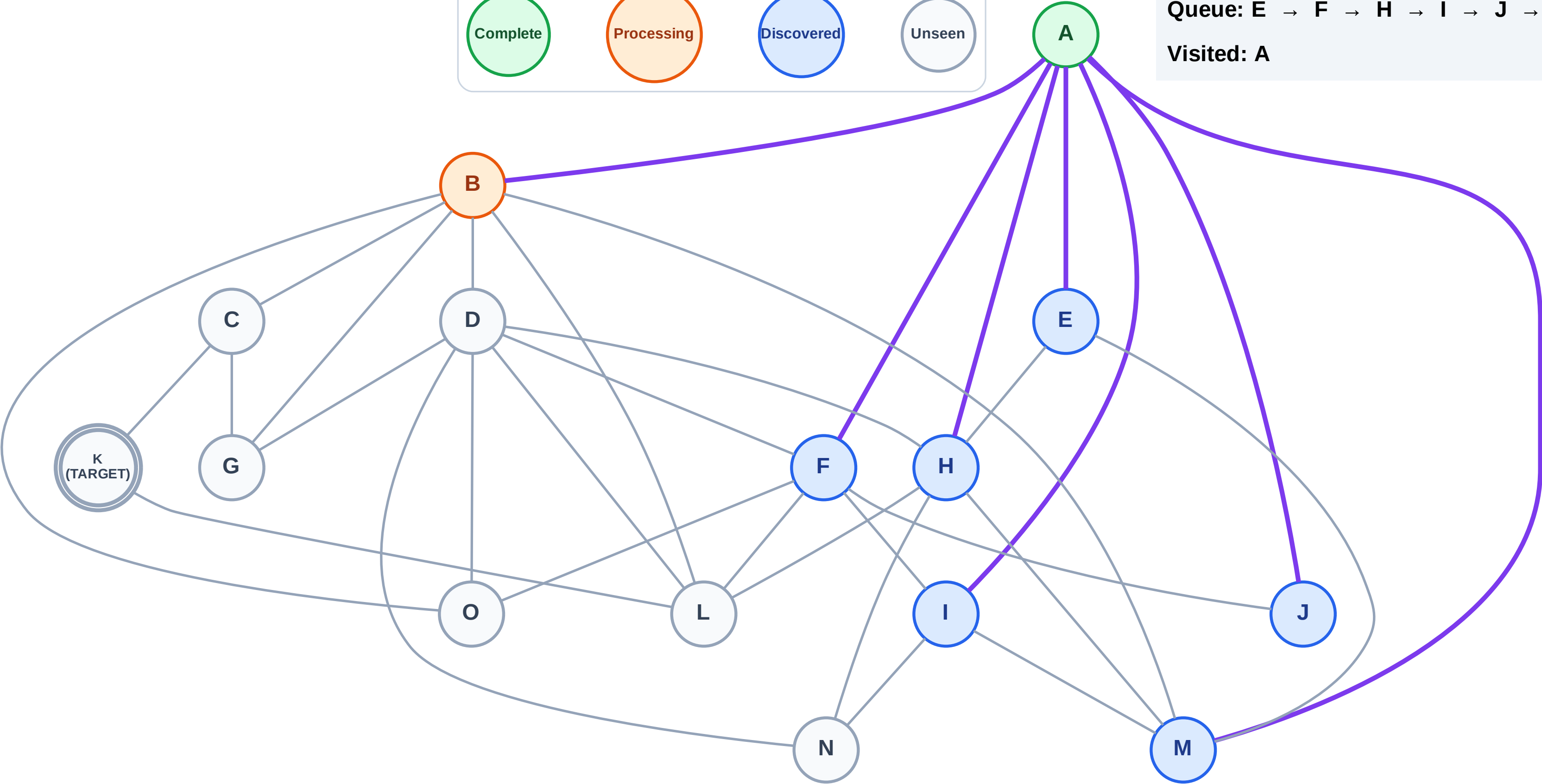
Processing

Discovered

Unseen

Queue: E → F → H → I → J → M

Visited: A



BFS Traversal

Step 5: Discover C, D, G, L, O from B

Legend



Complete



Processing



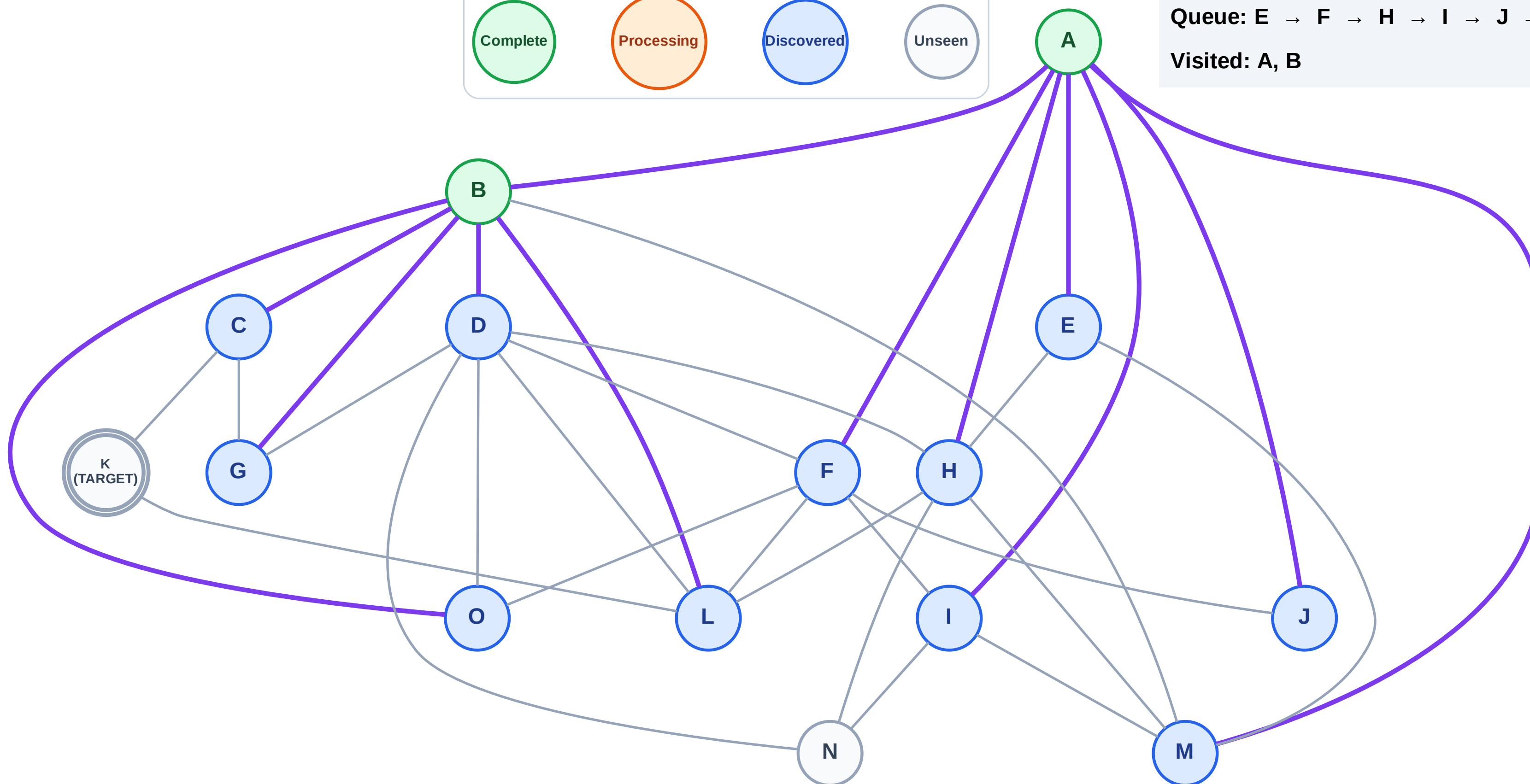
Discovered



Unseen

Queue: E → F → H → I → J → M → C → D → G → L → O

Visited: A, B



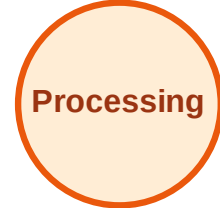
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



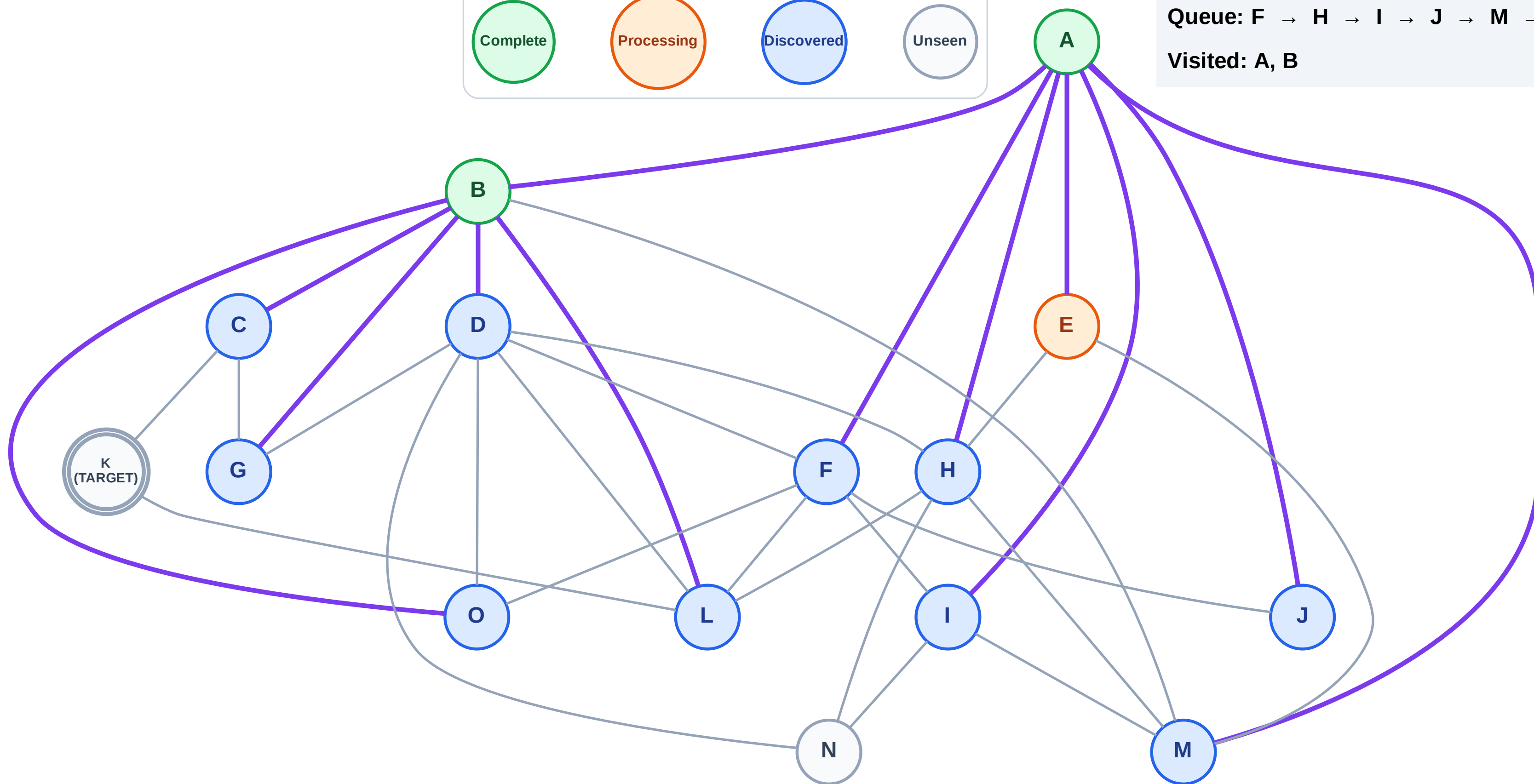
Discovered



Unseen

Queue: F → H → I → J → M → C → D → G → L → O

Visited: A, B



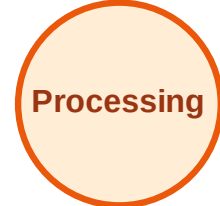
BFS Traversal

Step 7: No new neighbors from E

Legend



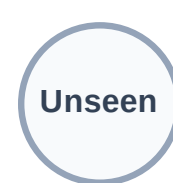
Complete



Processing



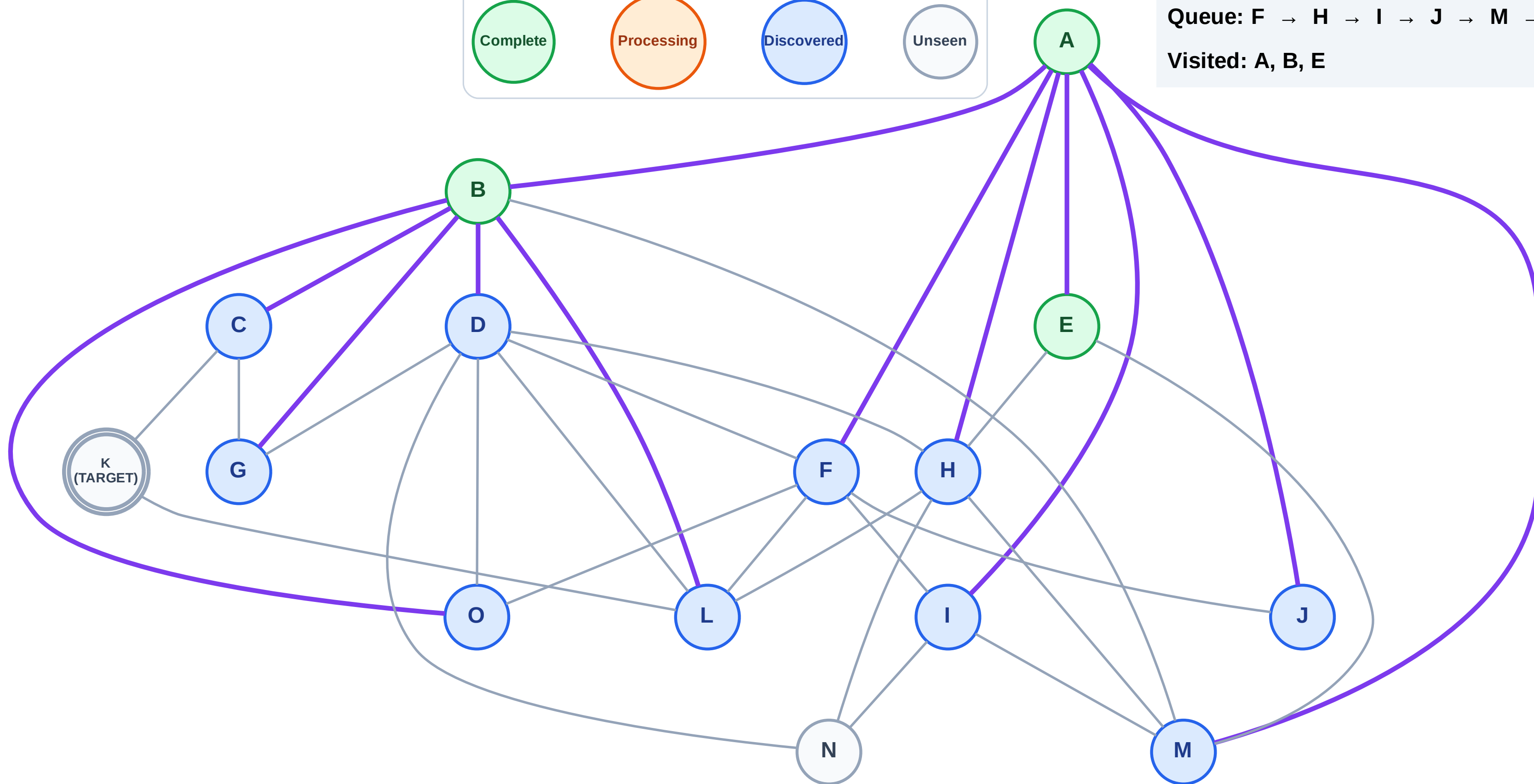
Discovered



Unseen

Queue: F → H → I → J → M → C → D → G → L → O

Visited: A, B, E



BFS Traversal

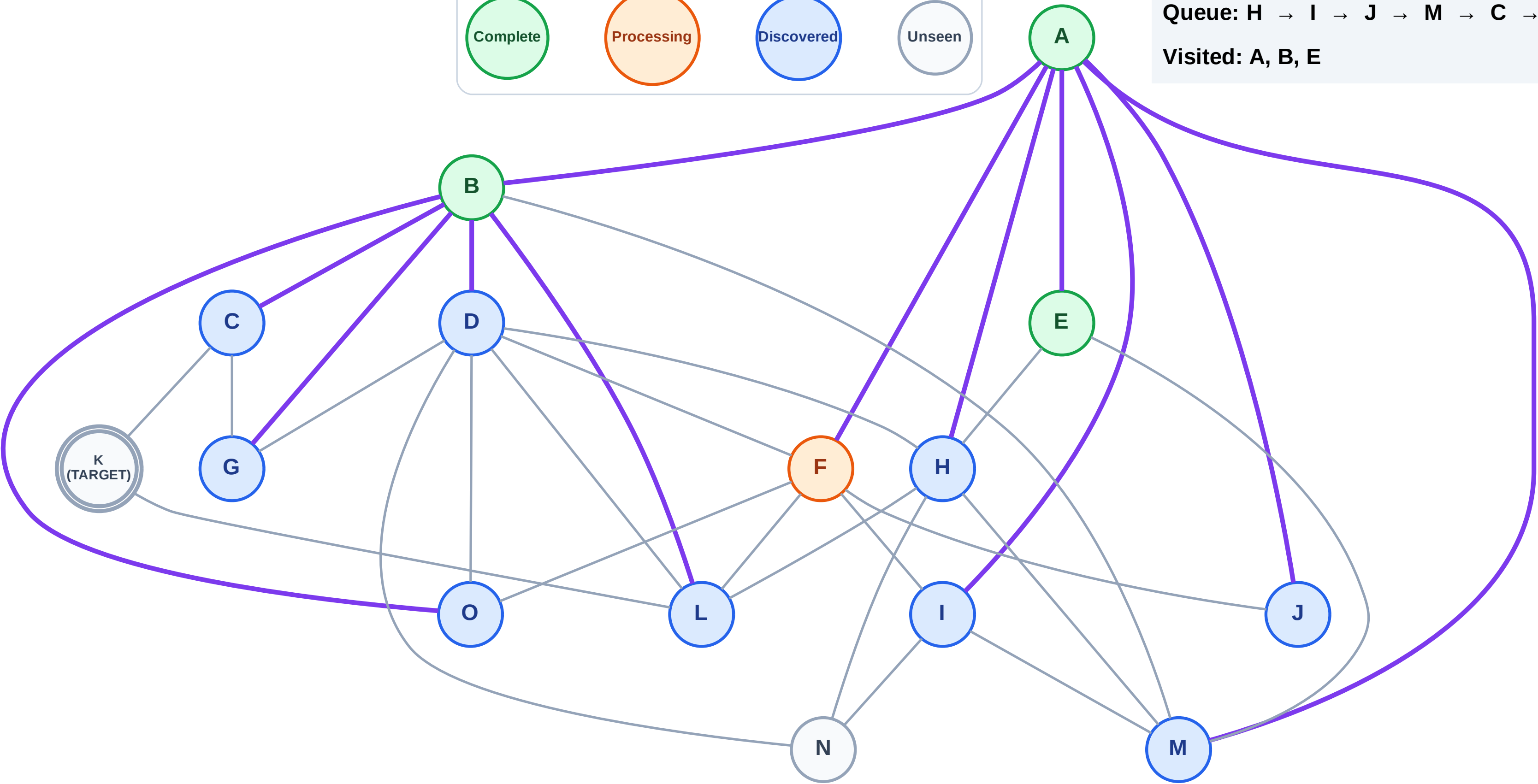
Step 8: Process node F

Legend



Queue: H → I → J → M → C → D → G → L → O

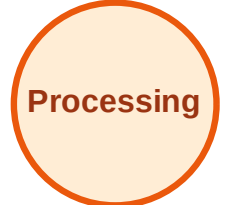
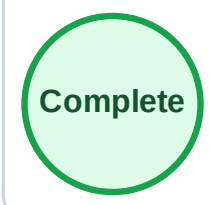
Visited: A, B, E



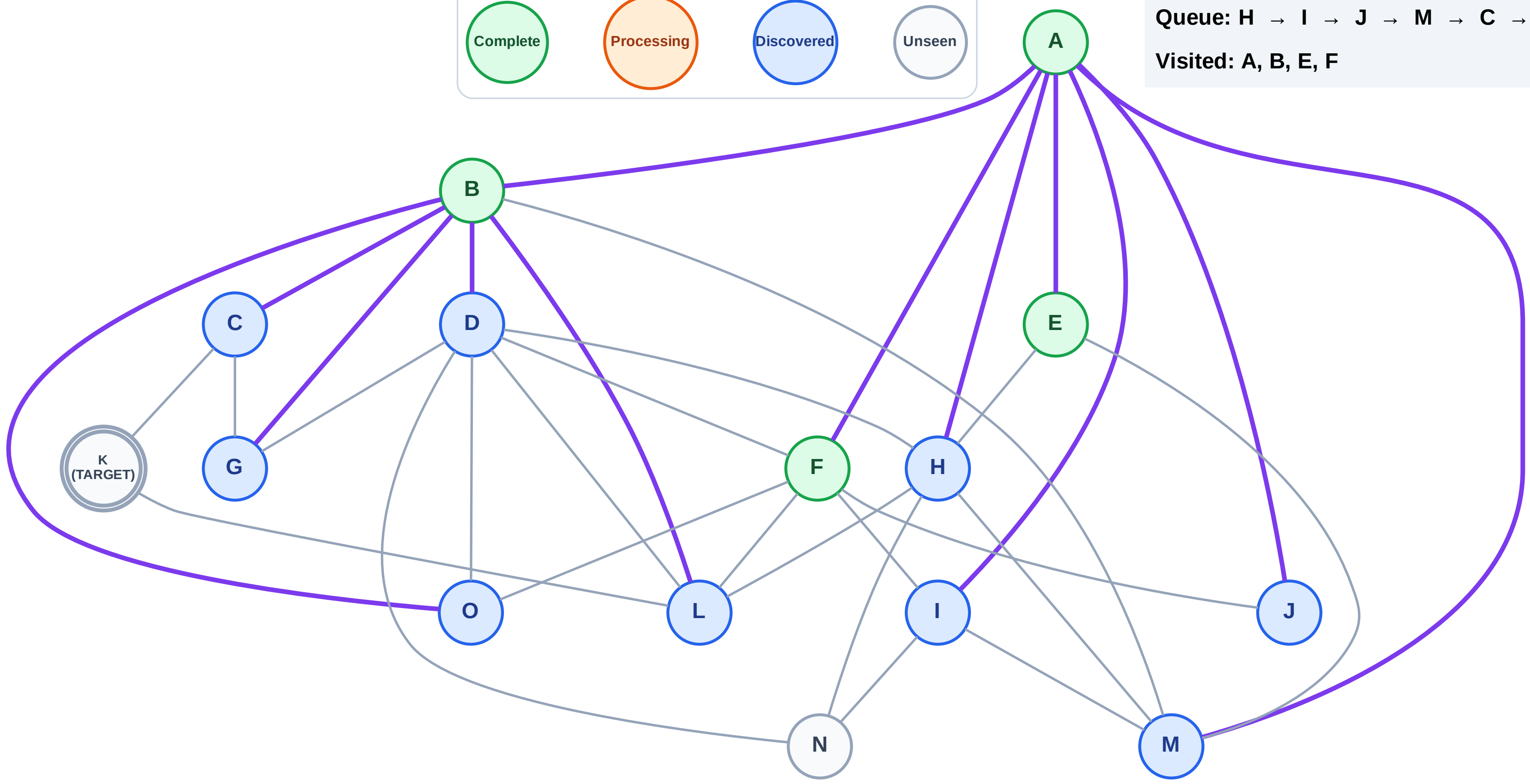
BFS Traversal

Step 9: No new neighbors from F

Legend



Queue: H → I → J → M → C → D → G → L → O
Visited: A, B, E, F



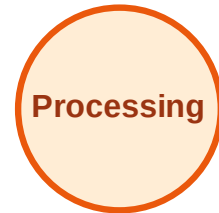
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



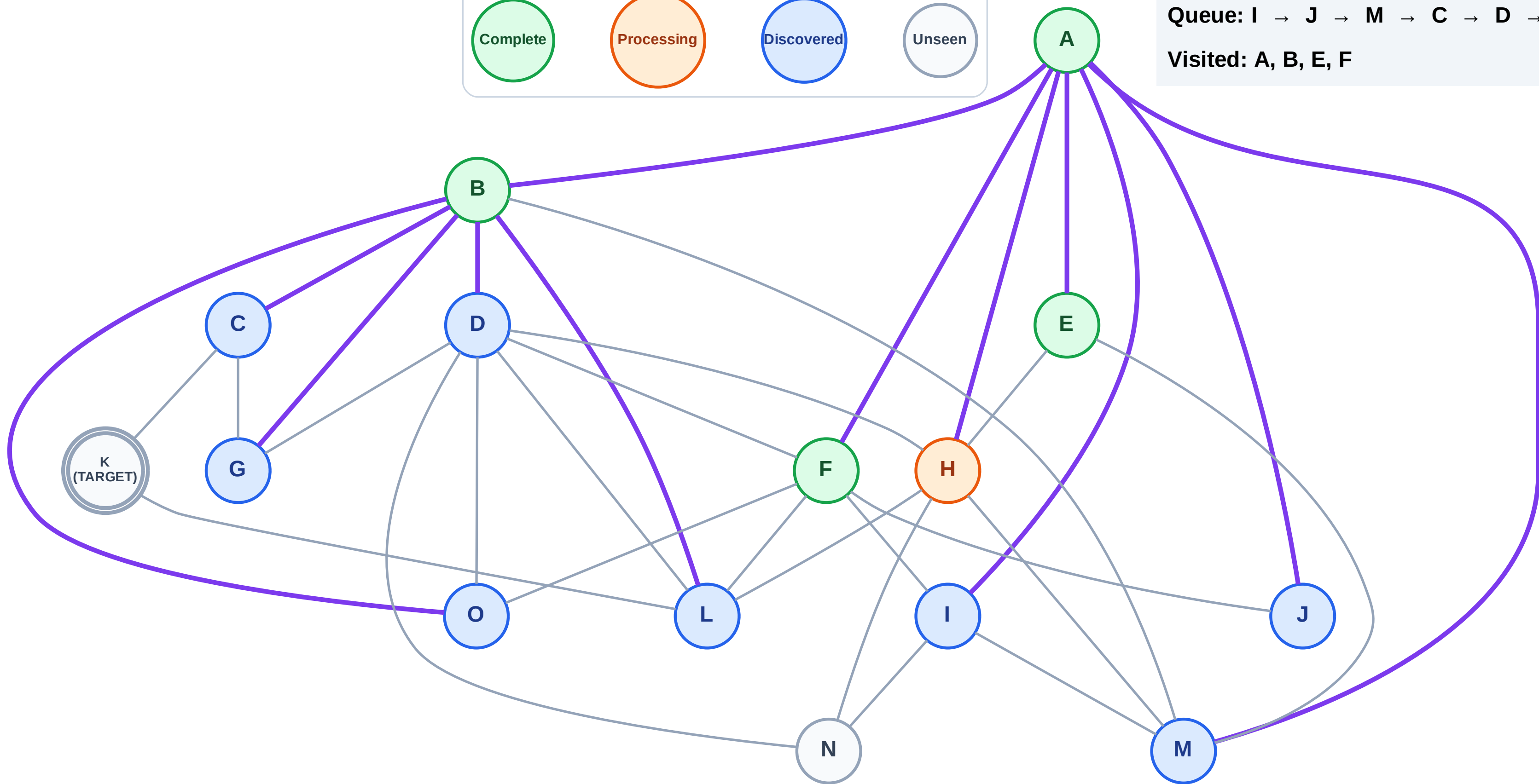
Discovered



Unseen

Queue: I → J → M → C → D → G → L → O

Visited: A, B, E, F



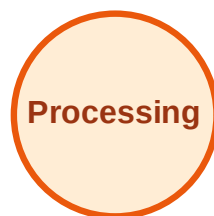
BFS Traversal

Step 11: Discover N from H

Legend



Complete



Processing



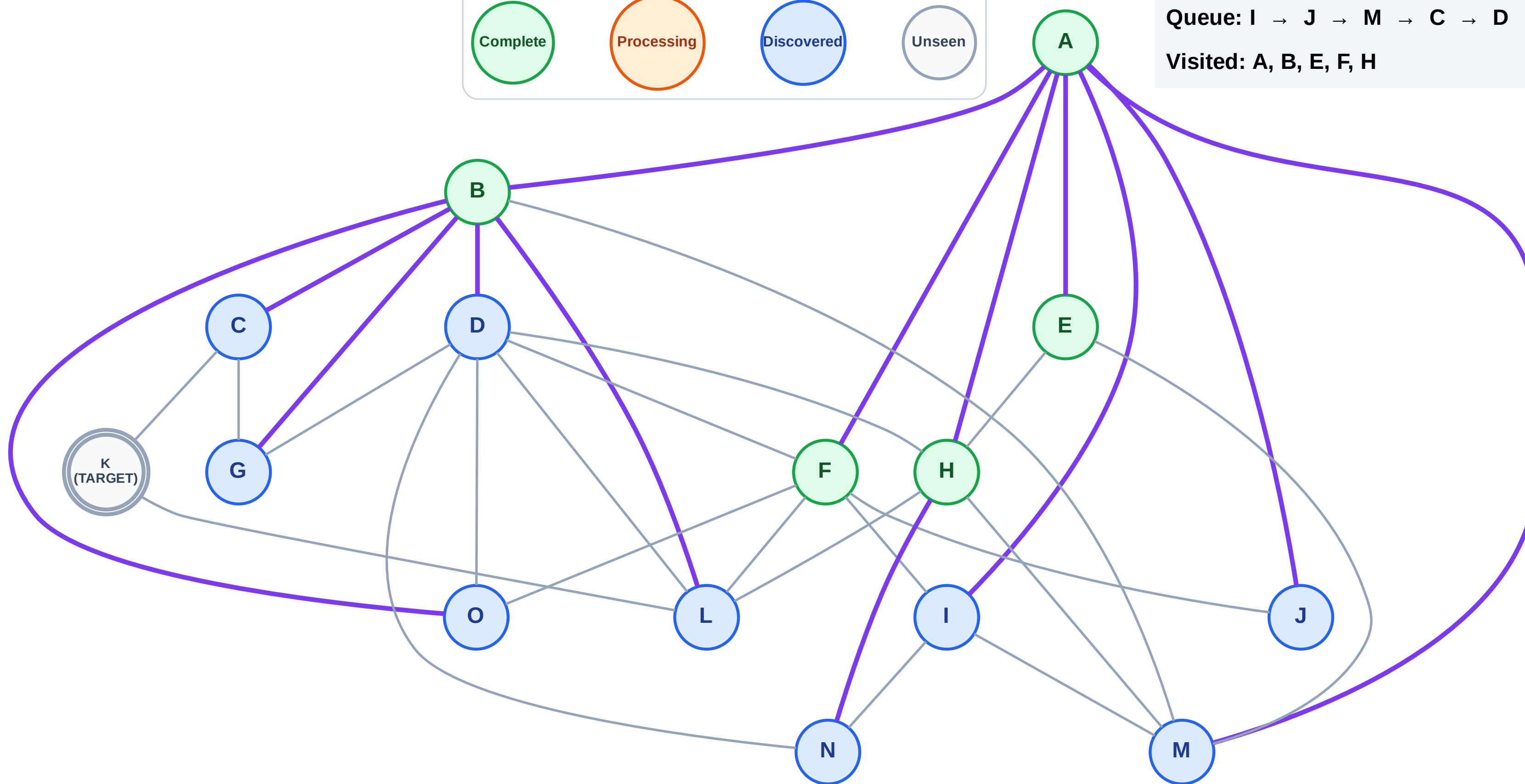
Discovered



Unseen

Queue: I → J → M → C → D → G → L → O → N

Visited: A, B, E, F, H



BFS Traversal

Step 12: Process node I

Legend

Complete

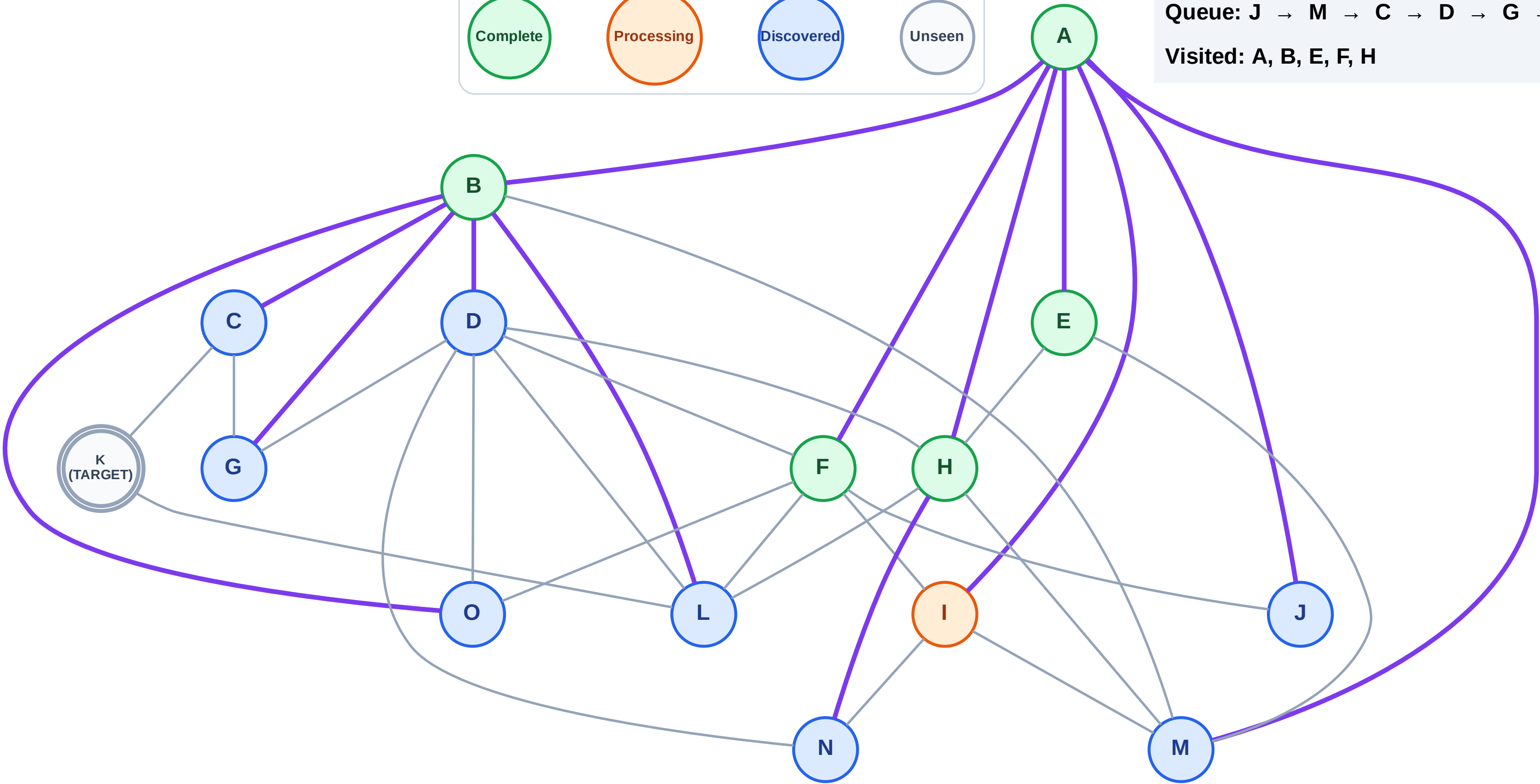
Processing

Discovered

Unseen

Queue: J → M → C → D → G → L → O → N

Visited: A, B, E, F, H



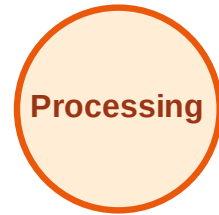
BFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing



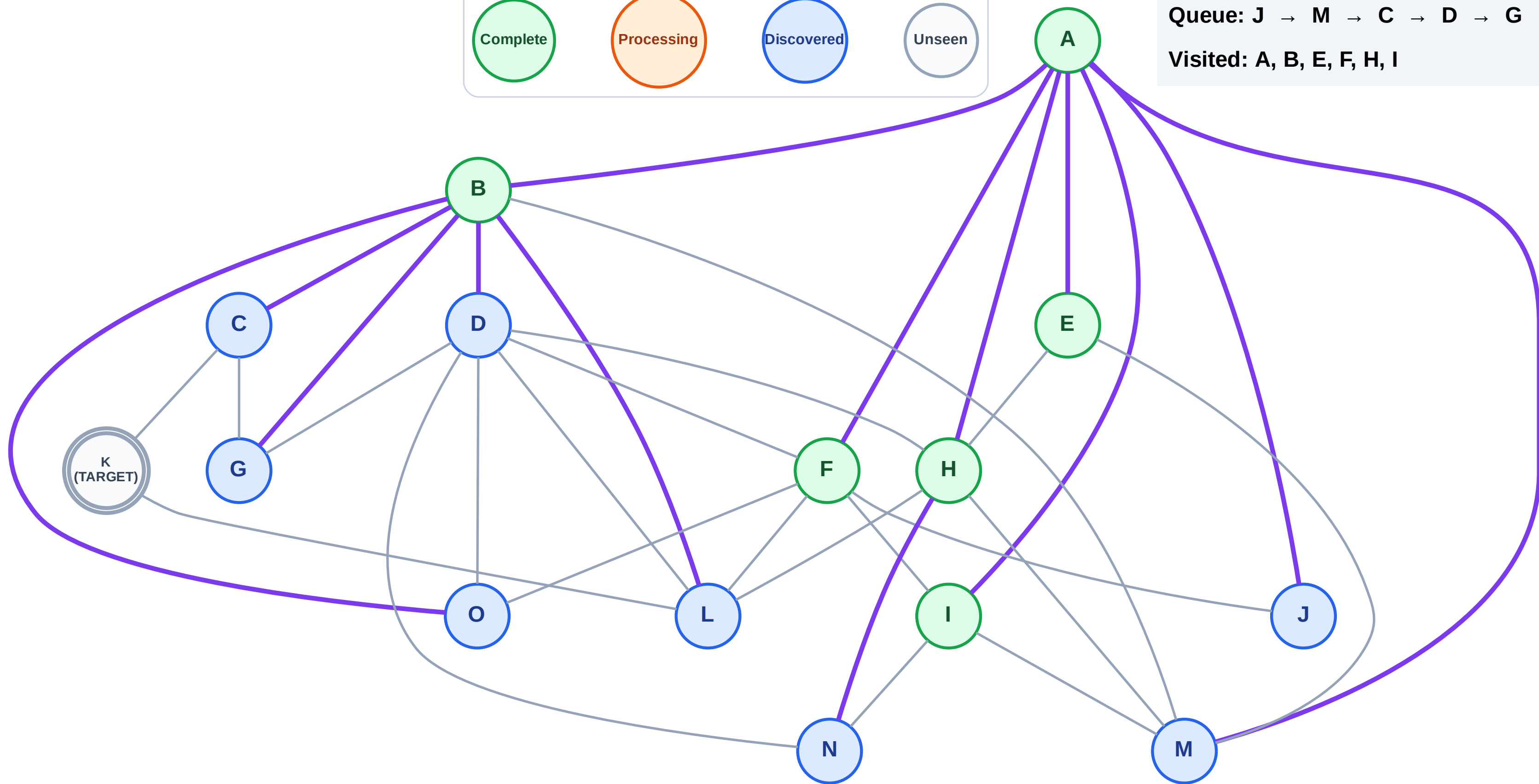
Discovered



Unseen

Queue: J → M → C → D → G → L → O → N

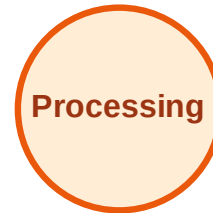
Visited: A, B, E, F, H, I



BFS Traversal

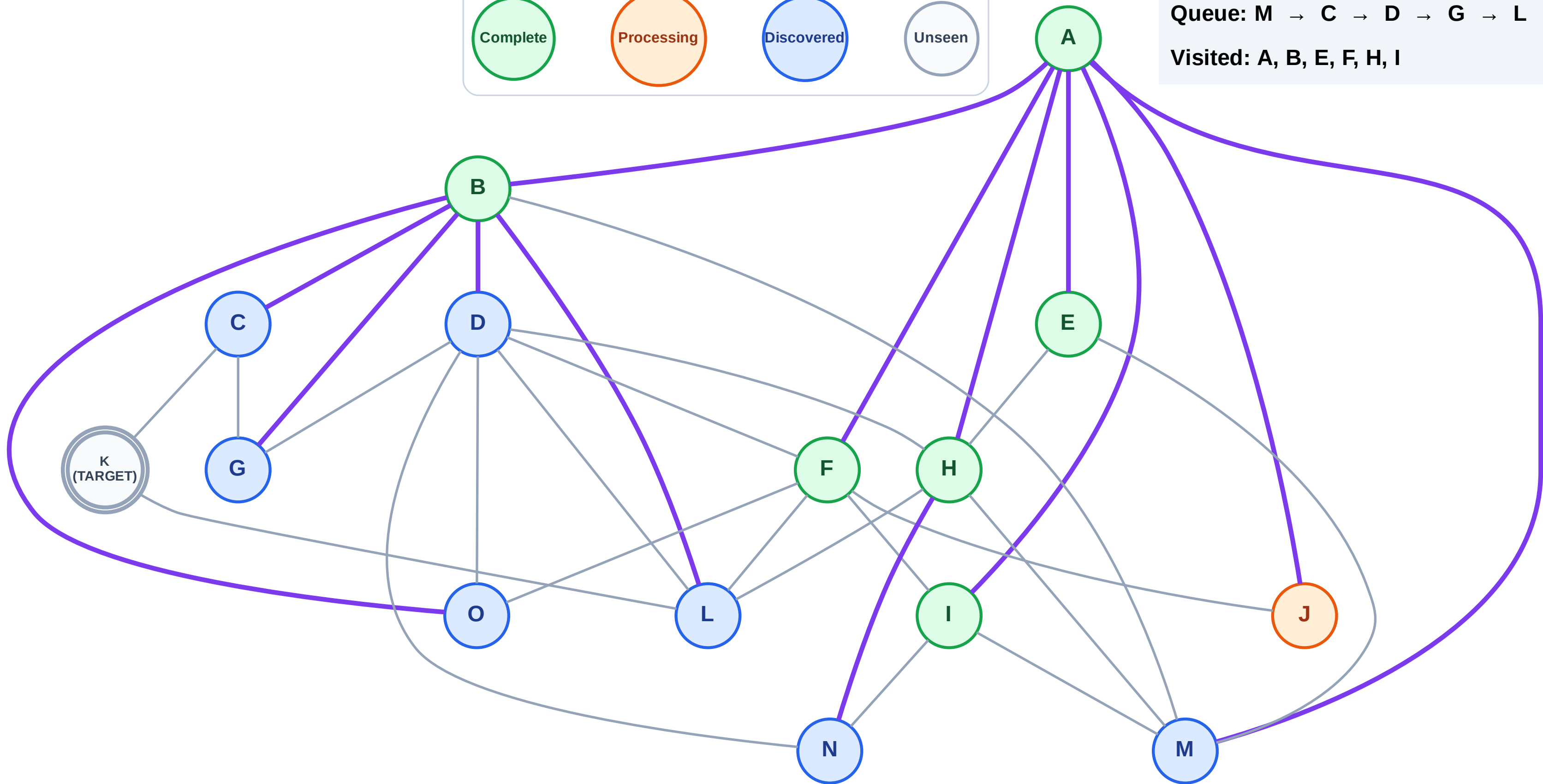
Step 14: Process node J

Legend



Queue: M → C → D → G → L → O → N

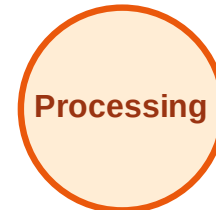
Visited: A, B, E, F, H, I



BFS Traversal

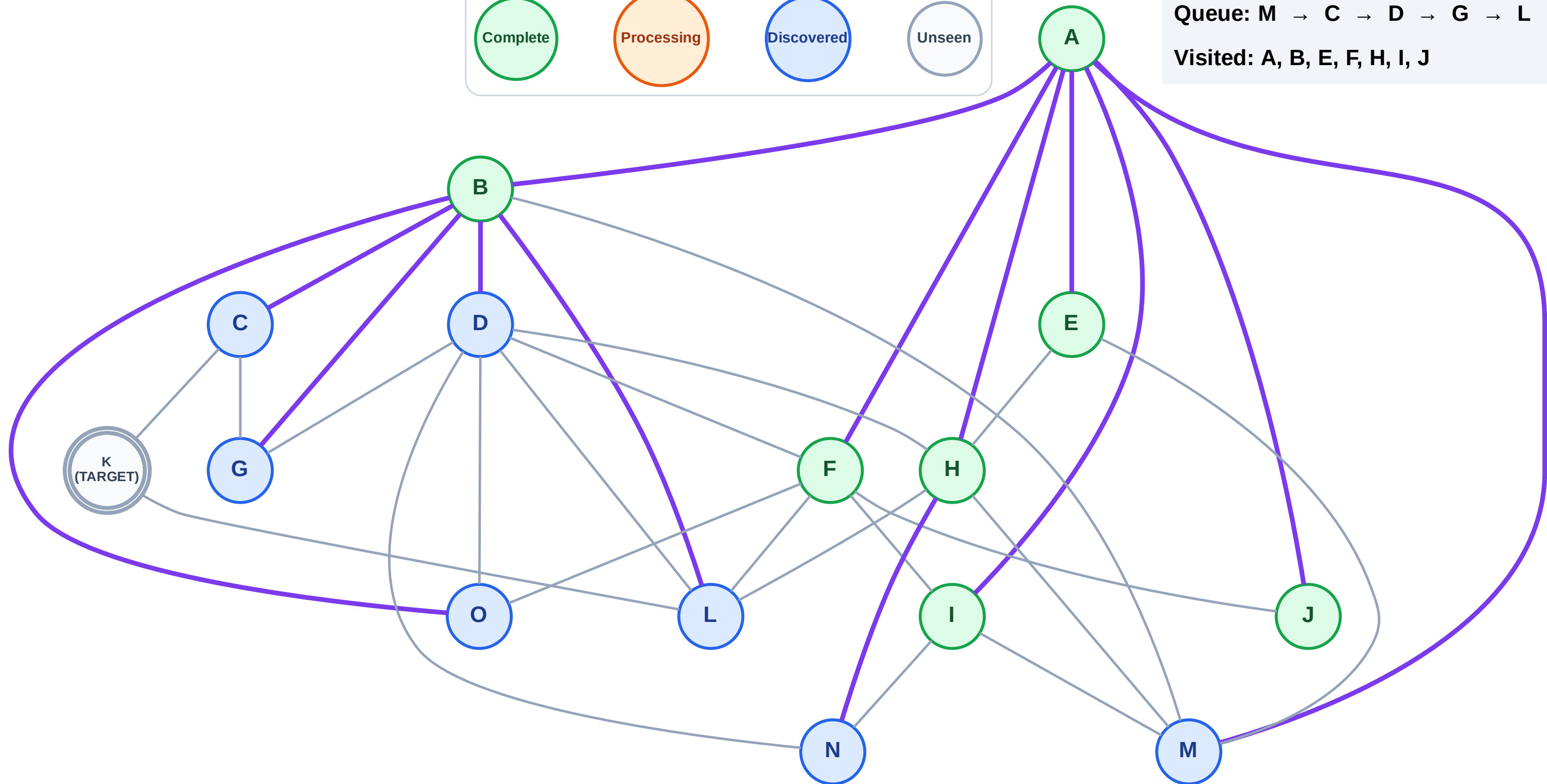
Step 15: No new neighbors from J

Legend



Queue: M → C → D → G → L → O → N

Visited: A, B, E, F, H, I, J



BFS Traversal

Step 16: Process node M

Legend

Complete

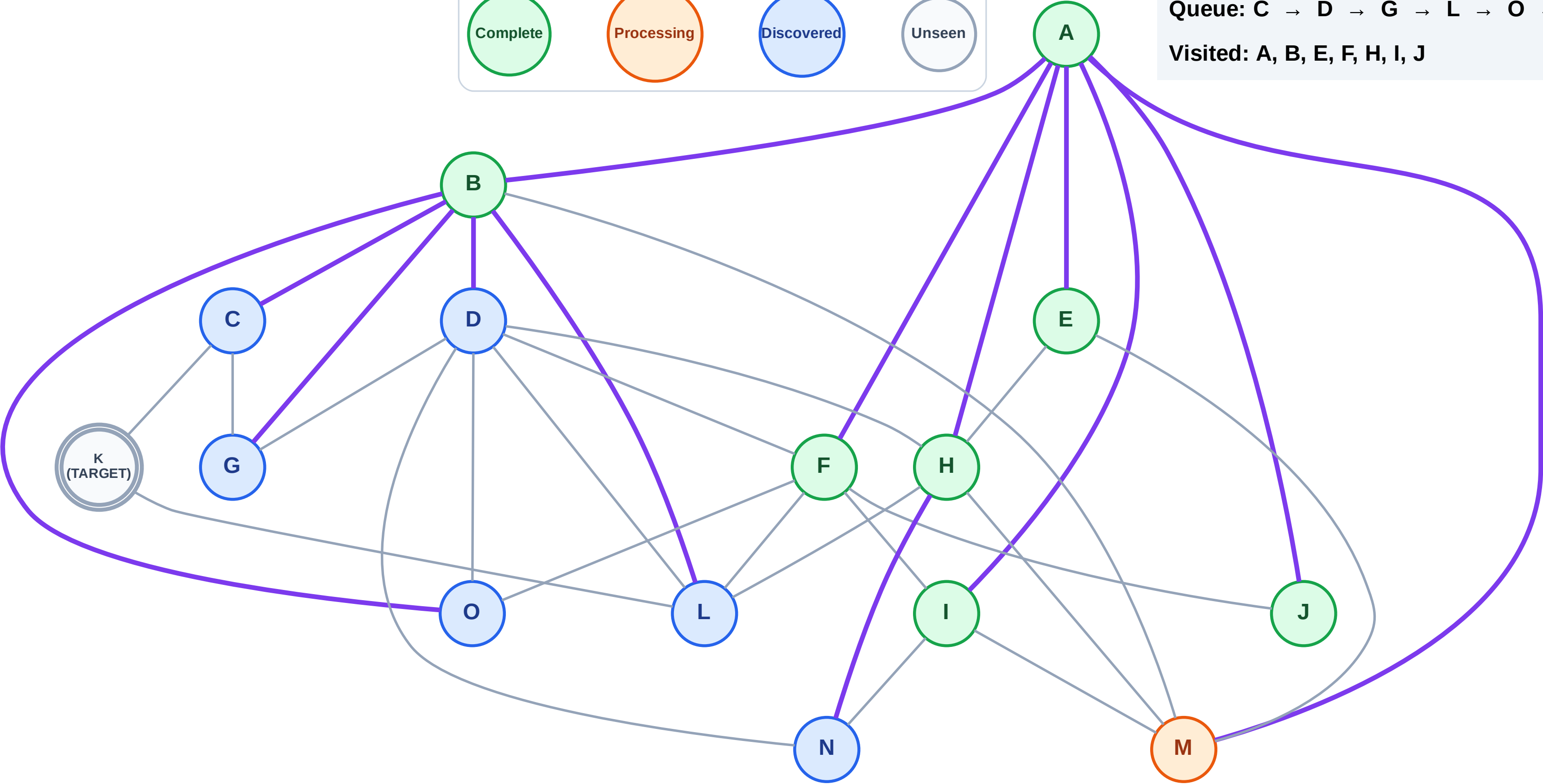
Processing

Discovered

Unseen

Queue: C → D → G → L → O → N

Visited: A, B, E, F, H, I, J



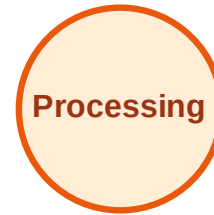
BFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing



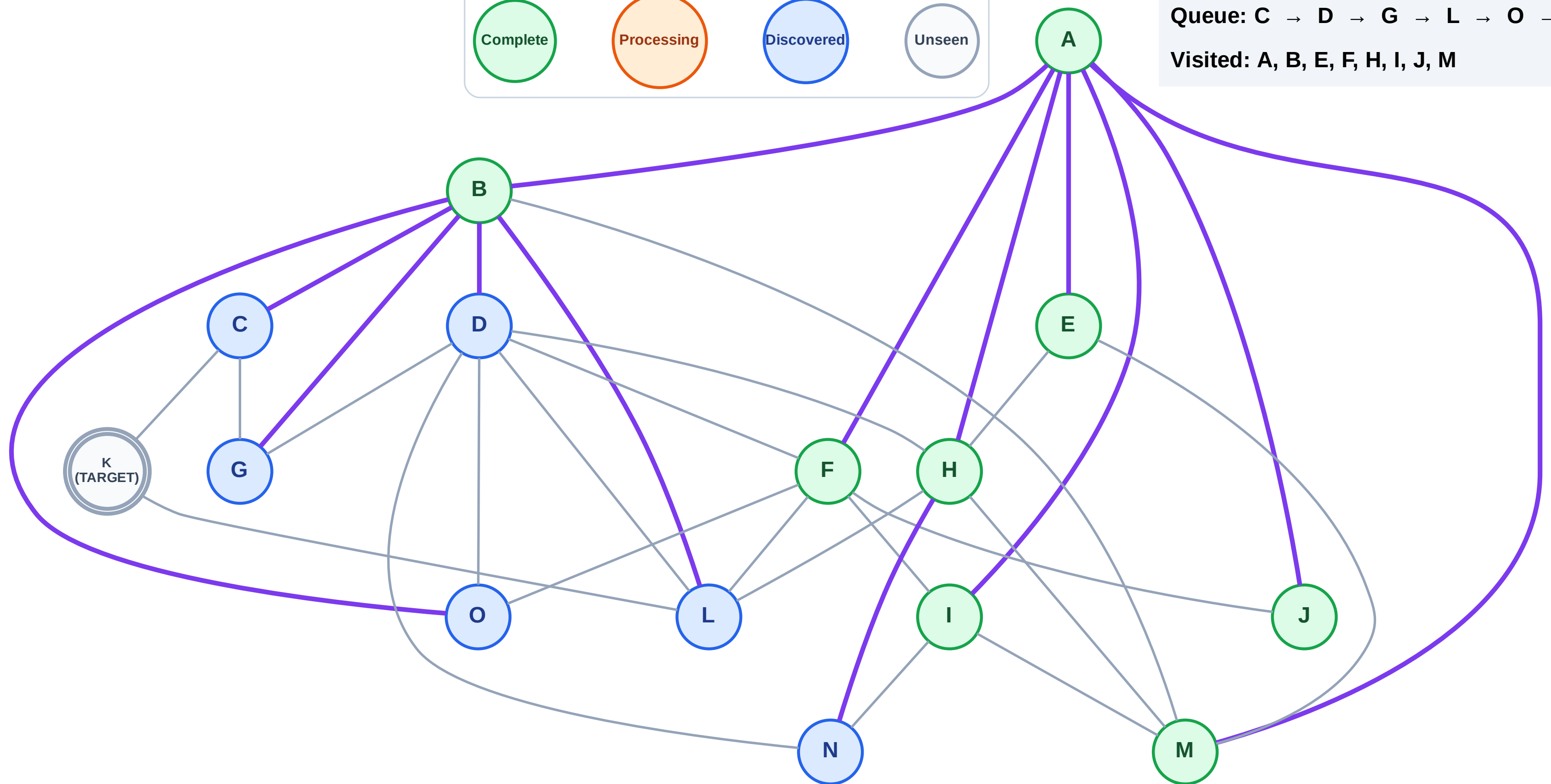
Discovered



Unseen

Queue: C → D → G → L → O → N

Visited: A, B, E, F, H, I, J, M



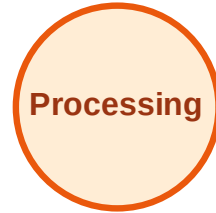
BFS Traversal

Step 18: Process node C

Legend



Complete



Processing



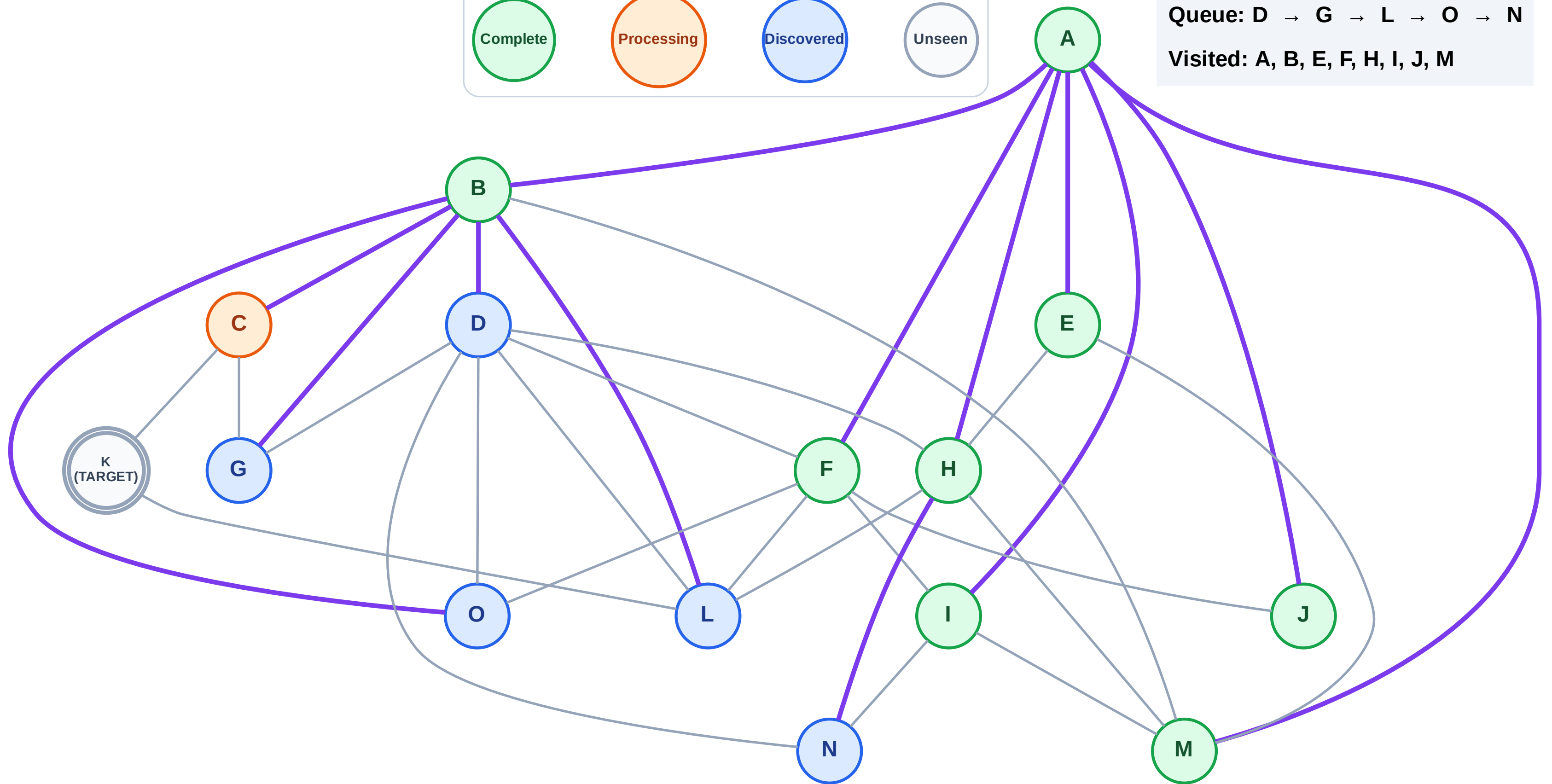
Discovered



Unseen

Queue: D → G → L → O → N

Visited: A, B, E, F, H, I, J, M



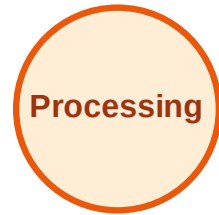
BFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Complete



Processing



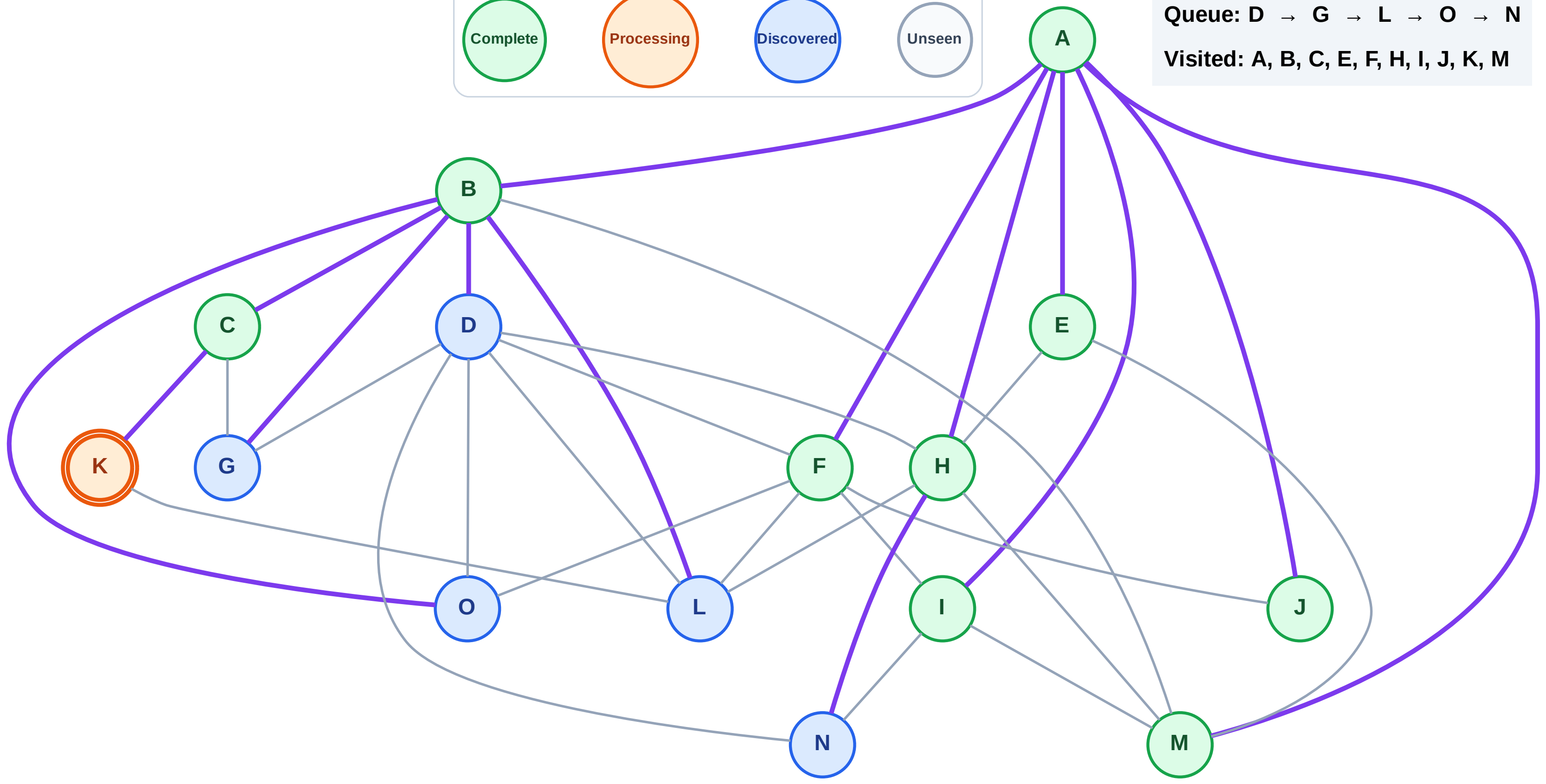
Discovered



Unseen

Queue: D → G → L → O → N

Visited: A, B, C, E, F, H, I, J, K, M



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

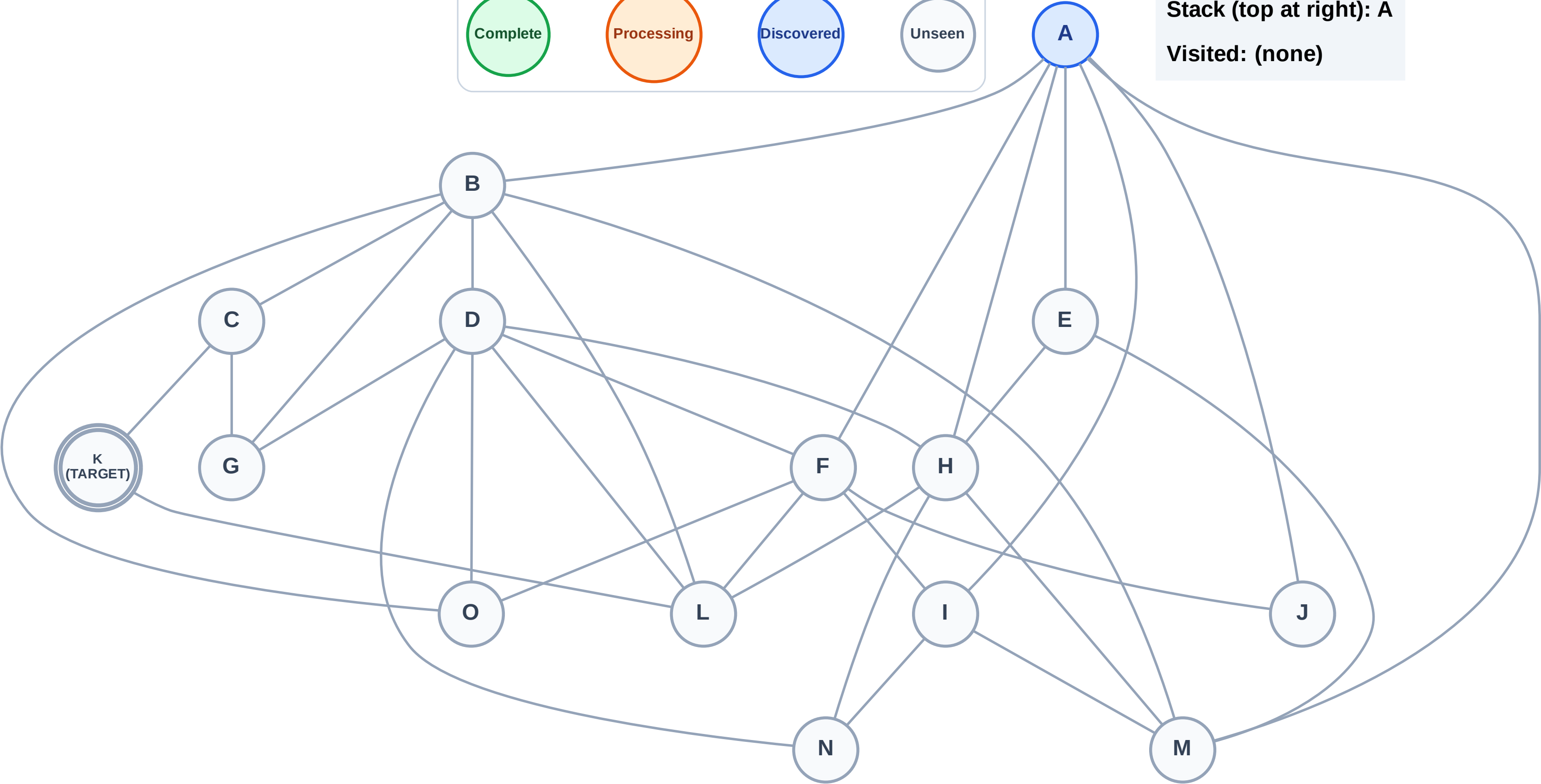
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

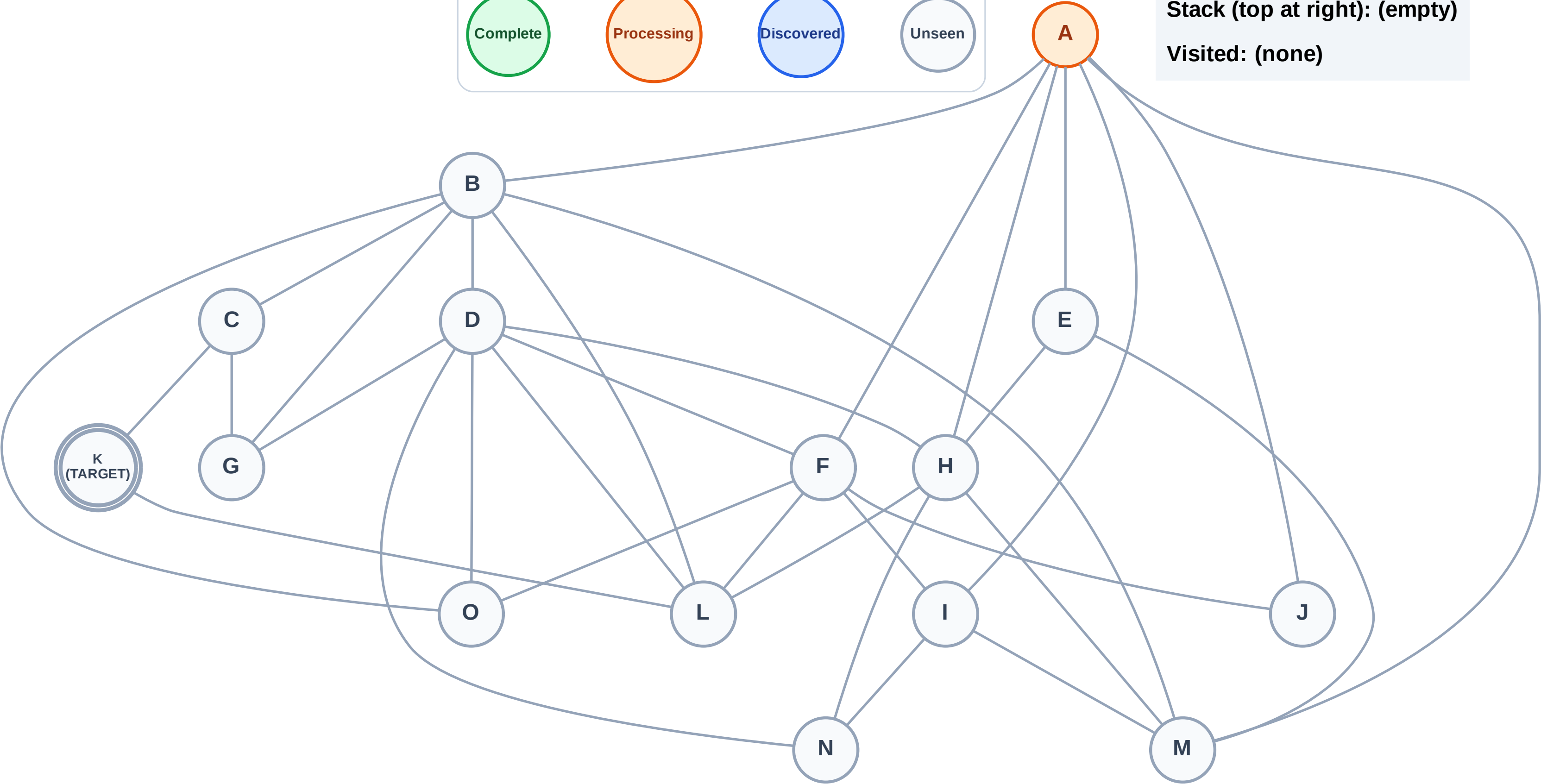
Complete

Processing

Discovered

Unseen

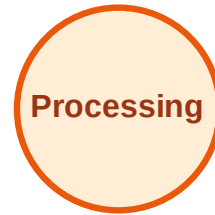
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

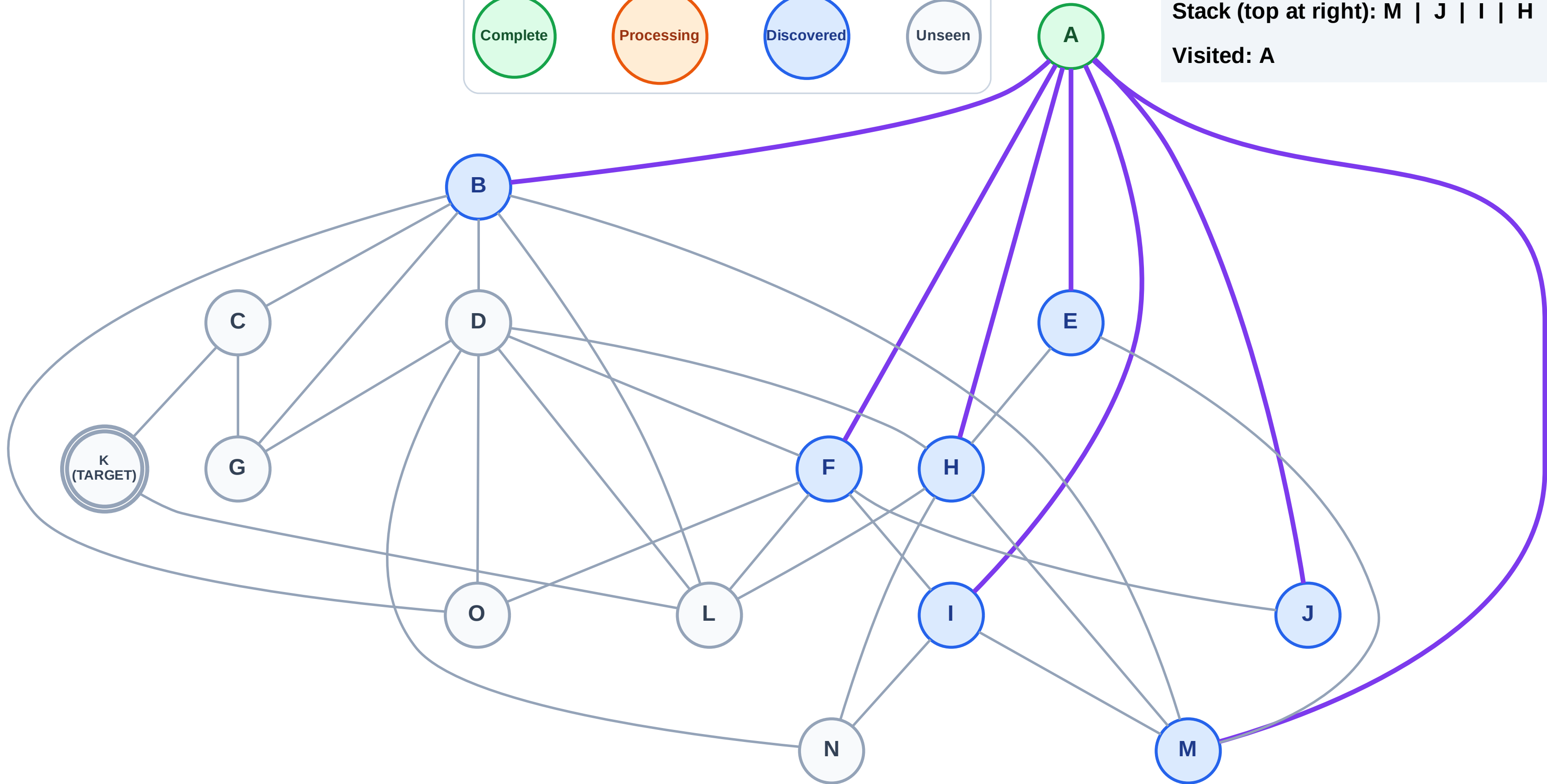
Step 3: Discover M, J, I, H, F, E, B from A

Legend



Stack (top at right): M | J | I | H | F | E | B

Visited: A



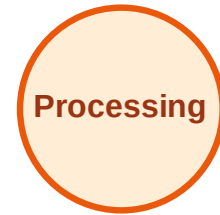
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



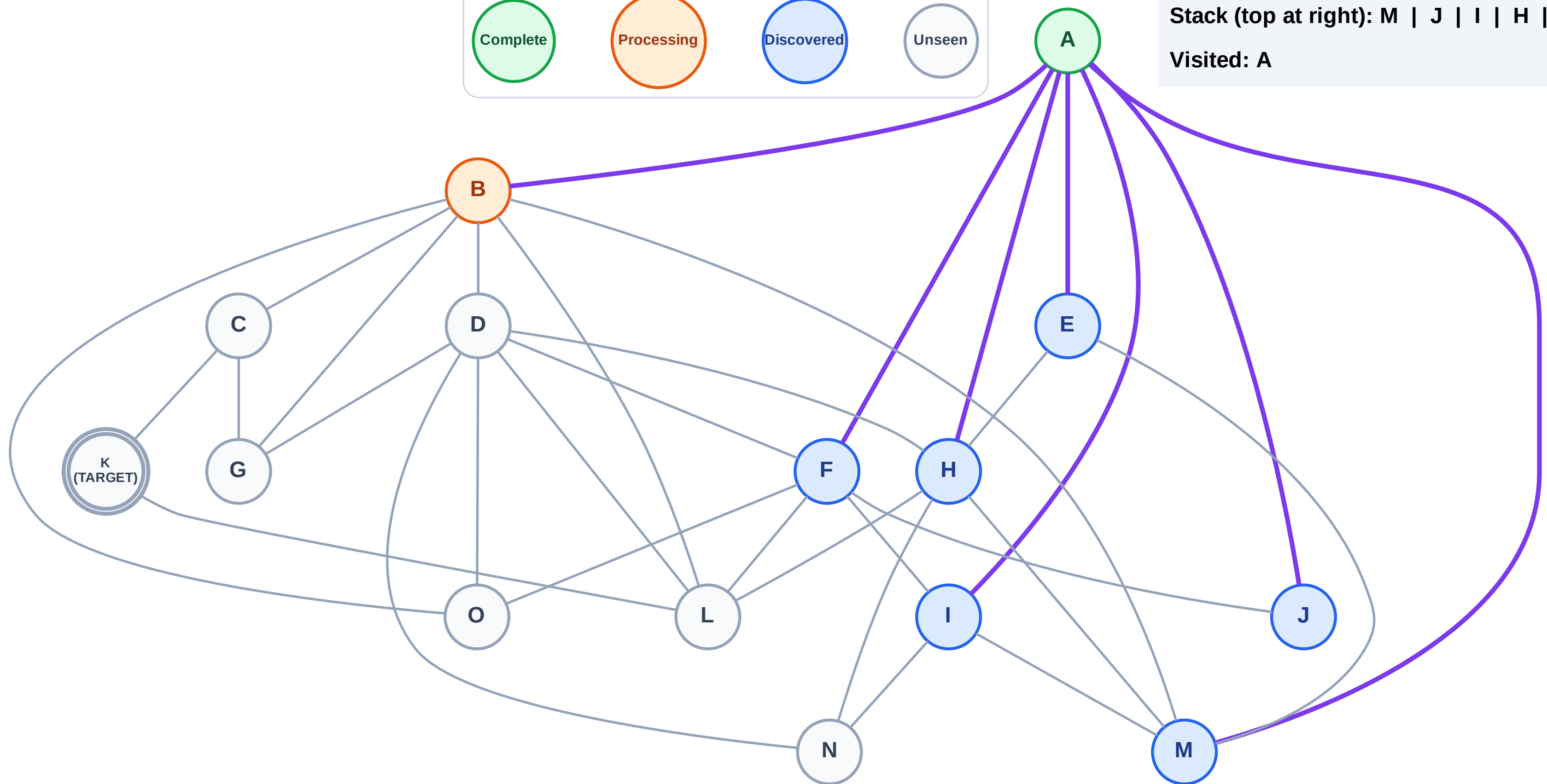
Discovered



Unseen

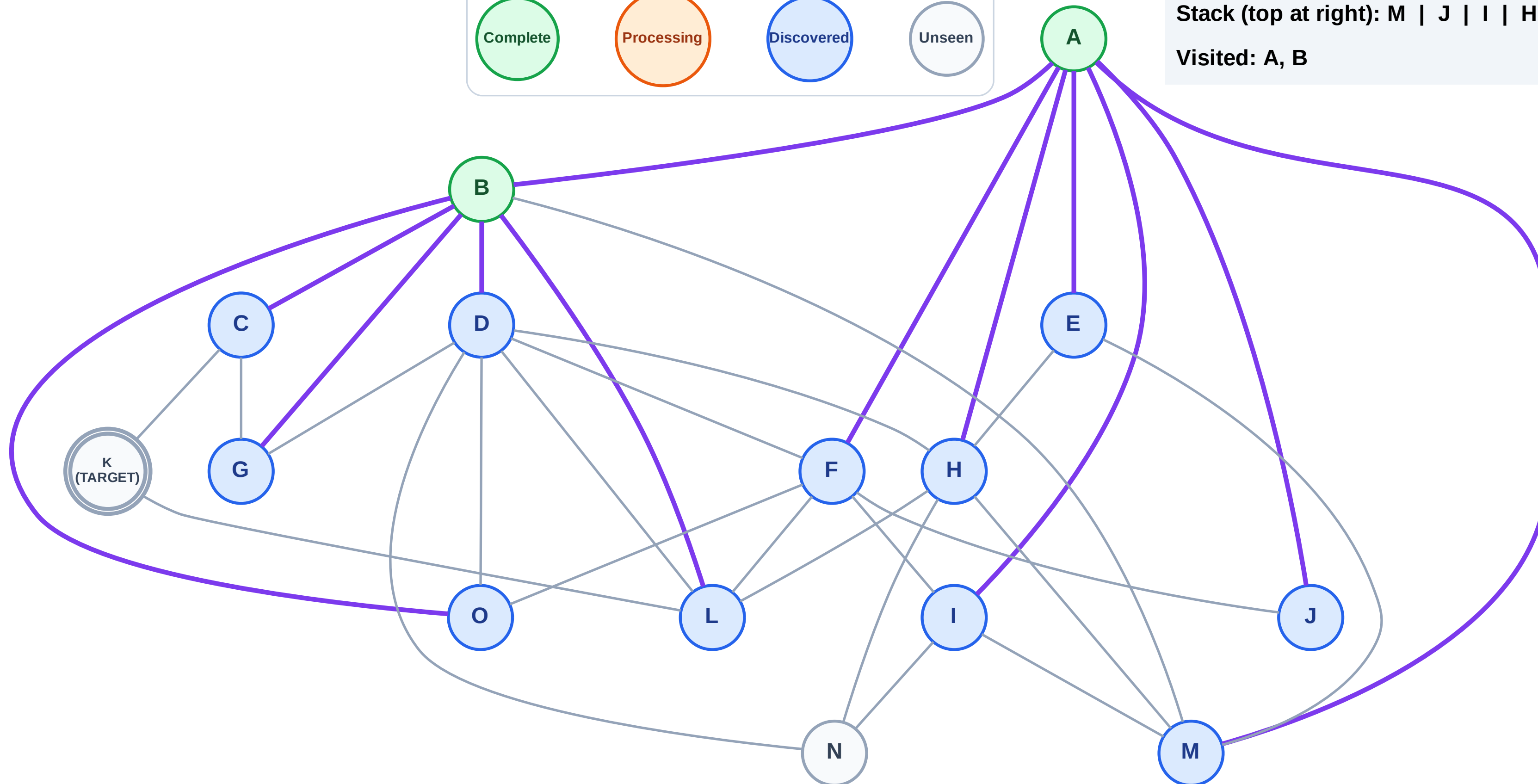
Stack (top at right): M | J | I | H | F | E

Visited: A



Step 5: Discover O, L, G, D, C from B

Step 5: Discover O, L, G, D, C from B



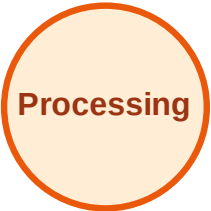
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



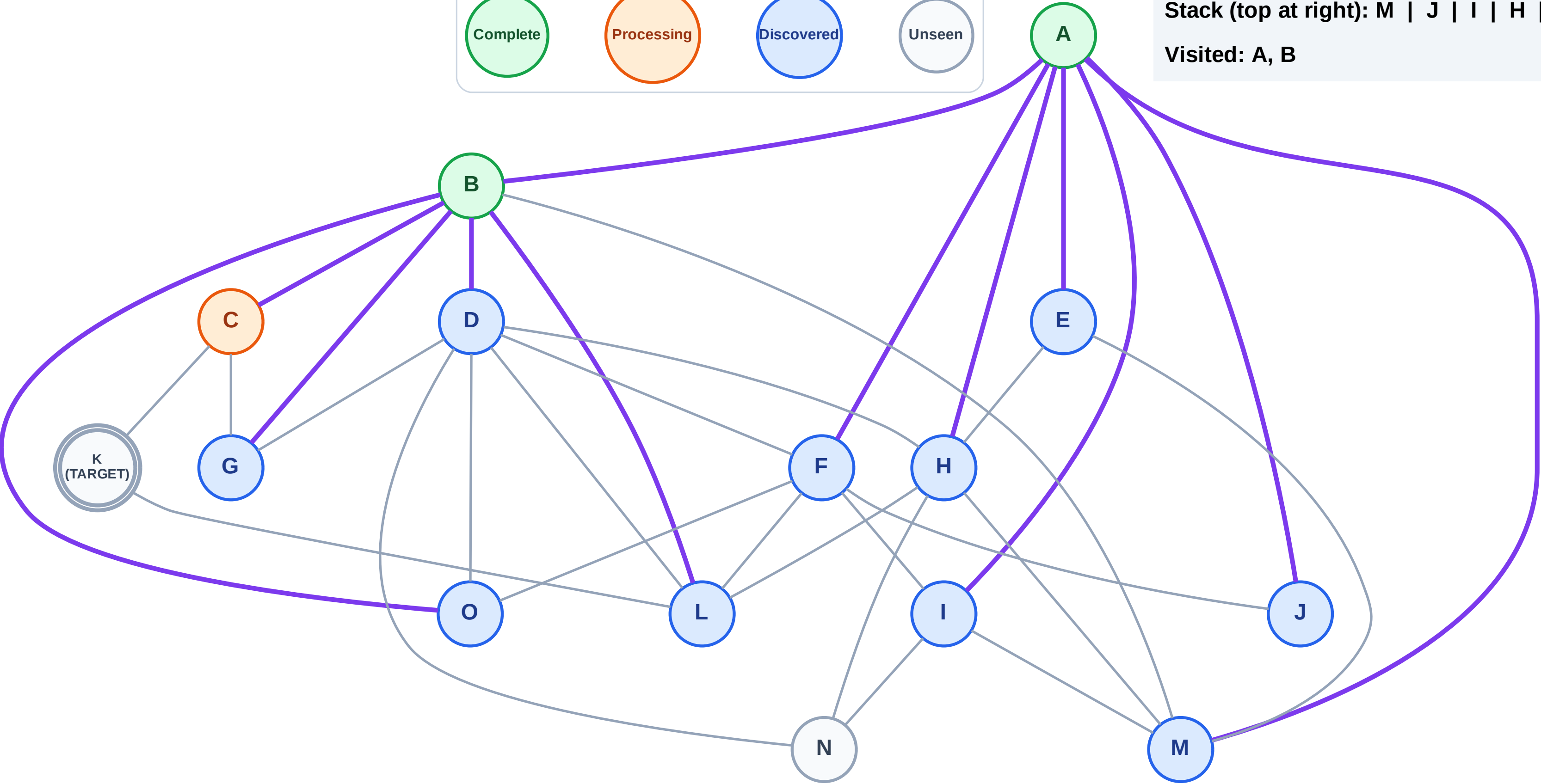
Discovered



Unseen

Stack (top at right): M | J | I | H | F | E | O | L | G | D

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Stack (top at right): M | J | I | H | F | E | O | L | G | D
Visited: A, B, C, K

